EXPLORATORY DATA ANALYSIS

# Chapter 1 — What Is EDA?

From Raw Data to Understanding

**Course focus**
Quantitative EDA and Visual EDA, taught together.

## Tools used throughout the course

Python • NumPy • Pandas • Matplotlib (object-oriented API) • Seaborn

Chapter notes, reflection questions, and a practical notebook template accompany this PDF.

# Chapter roadmap

This is the first chapter, so there are no earlier concepts, assignments, or coding projects to review.

### Learning objectives

• Define Exploratory Data Analysis (EDA) and explain its purpose.

• Distinguish quantitative EDA from visual EDA and explain why both matter.

• Use context and data quality questions before interpreting patterns.

• Describe a repeatable EDA workflow and recognize common analytical mistakes.

# 1. What is Exploratory Data Analysis?

EDA is the process of investigating a dataset by inspecting its structure, summarizing its values, visualizing patterns, and asking questions before drawing conclusions or building predictive models.

> **Central idea:** EDA is not merely a collection of plots or summary statistics. It is an iterative investigation whose goal is to understand what the data can—and cannot—tell you.

Think of entering an unfamiliar cave. Before deciding what it means, you look around, map its structure, note unusual features, and gather evidence. EDA plays a similar role when the unfamiliar environment is a dataset.

### A small example

Suppose a table contains age and salary:

| Age | Salary |
|---|---|
| 23 | 45,000 |
| 25 | 50,000 |
| 26 | 51,000 |
| 999 | 70,000 |

A model could accept age 999 as a number, but an analyst should question it. It might be a typo, a placeholder code, or a legitimate value in a badly misunderstood field. EDA raises the question; domain context and source investigation help answer it.

# 2. Three fundamental questions

### What do I have?

• How many rows and columns are present?

• What does each row represent?

• What do the columns mean, and what are their units?

• How were the observations collected?

### Can I trust the data?

• Are values missing, duplicated, or malformed?

• Are there impossible values or inconsistent labels?

• Do ranges and units make sense in context?

• Could the data collection process systematically omit certain observations?

### What patterns does the data contain?

• How are values distributed?

• Do groups differ?

• Which variables move together?

• Are there trends, clusters, gaps, or unusual observations?

## 3. Context comes before computation

A dataset is a record of some process—not the process itself. To interpret it, understand its source, purpose, population, sampling method, time period, measurement units, and meaning of one observation. A row might represent a person, transaction, visit, product, or daily summary. Confusing these units can produce misleading counts and comparisons.

> **Habit:** Before asking what a statistic means, ask what was measured, how it was measured, and which population the data represents.

## 4. Quantitative EDA and Visual EDA

### Quantitative EDA

Quantitative EDA uses numerical summaries and calculations: counts, proportions, mean, median, quantiles, variance, standard deviation, ranges, missingness rates, and association measures. It is useful for precision, comparison, reproducibility, and detecting differences that are hard to estimate by eye.

### Visual EDA

Visual EDA uses plots such as histograms, boxplots, scatterplots, bar charts, and heatmaps. It helps reveal distribution shape, clusters, nonlinear relationships, gaps, and unusual observations that a single statistic can conceal.

### Why use both?

Consider heights 170, 171, 170, 169, 171, and 220. The mean is 178.5, but that one number does not explain why the average is high. A plot would make the unusually large value obvious. Then you would investigate whether 220 is a valid observation, an error, or a different population.

> **Course principle:** Neither numbers nor plots are universally superior. Choose the method based on the question, and use the other as a complementary check.

## 5. Observation is not explanation

Suppose ice-cream sales and shark incidents both rise during summer. The data may show a positive association, but that does not establish that ice cream causes shark incidents. Warmer weather could increase both ice-cream purchases and beach attendance. EDA describes and investigates patterns; causal explanations require stronger designs and evidence.

Prefer careful language: “These variables are associated in this dataset” rather than “This variable causes that outcome.” Record alternative explanations and what evidence would distinguish them.

## 6. A repeatable EDA workflow

| Step | Main question |
|---|---|
| 1. Understand context | What process and population produced the data? |
| 2. Inspect structure | What do rows, columns, types, and units represent? |
| 3. Assess quality | What is missing, duplicated, invalid, or inconsistent? |
| 4. Quantify | What do counts, distributions, and summaries show? |
| 5. Visualize | What shapes, clusters, trends, or anomalies are visible? |
| 6. Study relationships | How do variables or groups relate? |
| 7. Investigate anomalies | What could explain unusual observations? |
| 8. Communicate | What did we learn, and what remains uncertain? |

The process is iterative. A plot may raise a question that sends you back to the data dictionary or source. Do not treat the workflow as a rigid one-way checklist.

# 7. Common beginner mistakes

• Starting model training before understanding the dataset.

• Relying only on plots or only on numerical summaries.

• Assuming collected data must be correct.

• Treating association as proof of causation.

• Ignoring units, collection methods, and the meaning of a row.

• Deleting unusual values before investigating them.

• Generating charts without a clear question or takeaway.

# 8. Chapter summary

• EDA is an iterative process for understanding data before drawing conclusions or modeling.

• The three core questions are: What do I have? Can I trust it? What patterns does it contain?

• Quantitative and visual methods complement one another.

• Context determines whether values and patterns make sense.

• Good analysts distinguish observation from explanation and communicate uncertainty.

# Assignment

## Part A — Reflection

Why is EDA performed before machine learning?

How can quantitative and visual EDA reveal different aspects of the same dataset?

Why should analysts avoid rushing to explanations?

What are the three fundamental questions of EDA?

Why is dataset context as important as its values?

## Part B — Thought exercise

Imagine a customer dataset with Customer ID, Age, Income, Occupation, City, and Purchased Product. Write at least 15 questions you would ask before analyzing it. Group them under context, structure, quality, distributions, and relationships.

### Part C — Mini coding exercise

Choose a small dataset or CSV. Using Python, NumPy, and Pandas, inspect its row and column count, column names, data types, missing values, duplicate rows, and first/last five rows. Do not make plots yet.

```
# Starter example (replace this with your chosen dataset)
import numpy as np
import pandas as pd

df = pd.read_csv("your_dataset.csv")

print("Shape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nData types:\n", df.dtypes)
print("\nMissing values:\n", df.isna().sum())
print("\nDuplicate rows:", df.duplicated().sum())
display(df.head())
display(df.tail())
```

## Preview of Next Chapter

### Chapter 2 — Understanding the Dataset

Next we will learn to read a dataset before analyzing it: observational units, rows and columns, data dictionaries, metadata, variable types, measurement scales, granularity, and the data-generating process. These foundations help prevent technically correct but conceptually wrong analyses.

Companion file: **EDA_Chapter_01_Notebook_Template.ipynb** — a ready-to-use notebook with prompts and starter code for the chapter's mini coding exercise.